

**A PROJECT TITLE
ON**

Hybrid Machine Learning Framework for Chronic Kidney Disease Prediction
project report submitted in partial fulfilment of the requirements for the Course

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Abstract

This project presents a hybrid machine learning framework for binary Chronic Kidney Disease (CKD) prediction using structured clinical attributes. The study uses the Chronic Kidney Disease dataset from the UCI Machine Learning Repository, containing 400 patient records, 24 predictor attributes, and a binary target indicating CKD or not CKD. The workflow covers data inspection, missing-value analysis, preprocessing, exploratory analysis, feature analysis, stratified train/test splitting, model development, hyperparameter tuning, evaluation, sanity checking, and model comparison. Numerical variables are treated using median imputation and standardization, while categorical variables are handled through most-frequent imputation and one-hot encoding within leakage-safe preprocessing pipelines. Nine algorithms are considered: Logistic Regression, Decision Tree, K-Nearest Neighbors, Support Vector Machine, Random Forest, AdaBoost, Gradient Boosting, XGBoost, and LightGBM. Hyperparameter tuning is performed using controlled randomized search, followed by evaluation on an untouched 20% test split. On the current 80-record test set, Random Forest, LightGBM, XGBoost, Gradient Boosting, and AdaBoost achieved 100% accuracy, while Decision Tree, Logistic Regression, SVM, and KNN achieved 98.75%, 97.50%, 97.50%, and 96.25%, respectively. These benchmark results are interpreted cautiously because the dataset is small and they do not constitute clinical validation.

Keywords: Chronic Kidney Disease, Machine Learning, Classification, Feature Selection, Hyperparameter Tuning, Explainable AI

1. Introduction

1.1 Purpose of the Project

Chronic Kidney Disease is a long-term condition in which kidney function is impaired or kidney damage persists over time. Early identification is important because CKD may progress without prominent symptoms and can be associated with serious complications. In practical healthcare settings, clinicians use demographic information, vital measurements, laboratory investigations, and related clinical indicators to assess kidney health. A machine learning system can process multiple heterogeneous attributes simultaneously and learn patterns associated with CKD classification.

The purpose of this project is to develop a systematic and reproducible machine learning framework that can classify a patient record as CKD or not CKD from structured clinical attributes. The project is intended as an academic decision-support prototype rather than a clinically validated diagnostic system. The main stakeholders are students, researchers, healthcare-technology developers, and future users of clinical decision-support systems who require a transparent workflow for preprocessing, model comparison, evaluation, and deployment-oriented experimentation.

1.2 Problem Statement

The project addresses a binary supervised classification problem: given 24 clinical and demographic attributes of a patient, predict whether the patient belongs to the CKD class or the not-CKD class. The expected output is a binary prediction corresponding to CKD or not CKD, together with accuracy, precision, recall, F1-score, ROC-AUC, and confusion-matrix based evaluation. The framework must handle missing values and mixed numerical/categorical attributes while preventing information leakage from the test set into model development.

1.3 Existing Methods and Disadvantages

Existing CKD machine learning studies have investigated individual classifiers, ensemble models, deep learning, explainable artificial intelligence, and web-based prediction systems. Recent research has shown that tree ensembles and boosting methods can perform strongly on structured CKD data, while explainability methods such as SHAP and LIME can improve the interpretability of predictions. However, direct comparison across studies is difficult because datasets, preprocessing strategies, validation procedures, target definitions, and evaluation metrics differ.

Several limitations remain important. Small benchmark datasets can produce optimistic results and may not represent diverse clinical populations. Missing values, class imbalance, feature-selection choices, and inappropriate validation can lead to leakage or overfitting. Some models also have limited interpretability, and high benchmark accuracy does not automatically imply clinical usefulness. These limitations motivate a controlled framework in which preprocessing is placed inside pipelines, the test set is kept untouched until final evaluation, multiple model families are compared, and the results are interpreted together with their generalization limitations.

1.4 Proposed System

The proposed system integrates data inspection, preprocessing, feature analysis, model development, controlled hyperparameter tuning, evaluation, sanity checking, and model comparison into one reproducible workflow. The study compares nine algorithms representing linear, distance-based, kernel, tree, bagging, and boosting approaches. Numerical predictors are median-imputed and standardized, categorical predictors are most-frequently imputed and one-hot encoded, and the target is encoded only for classification. A stratified 80:20 train/test split is used, while preprocessing is fitted only on training data. The framework is designed to support academic experimentation and later extension into a web-based prediction and dashboard interface.

1.5 Research Objectives

1. Prepare and validate a suitable CKD dataset for binary classification.
2. Perform systematic data preprocessing and exploratory data analysis.
3. Analyze and select relevant features while preventing data leakage.
4. Develop and train multiple machine learning algorithms representing different model families.
5. Perform controlled hyperparameter tuning and evaluate generalization using cross-validation and a held-out test set.
6. Compare model performance using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices, and discuss practical limitations.

2. Literature Survey

Five recent and directly relevant studies from 2023–2026 were reviewed. The selected studies cover machine-learning-based CKD prediction, diagnosis, explainability, and ensemble learning. The survey is intentionally concise to suit a normal B.Tech project report.

S.No.	Reference	Title	Bibliographic Details	Gap	Objective / Methods	Inference	Future Direction
1	Sanmarchi et al. (2023)	Predict, diagnose, and treat chronic kidney disease with machine learning: A systematic literature review	Journal of Nephrology, 36(4), 1101–1117. DOI: 10.1007/s40620-023-01573-4	Reviewed existing CKD ML studies and identified issues with heterogeneous evaluation and limited clinical validation.	Systematic literature review of 68 CKD ML studies.	ML shows promise for CKD prediction and diagnosis, but generalizability and validation remain important concerns.	Use diverse datasets, external validation, and interpretable models.
2	Islam et al. (2023)	Chronic kidney disease prediction based on machine learning algorithms	Journal of Pathology Informatics, 14, 100189. DOI: 10.1016/j.jpi.2023.100189	Need reliable CKD prediction using clinical attributes and suitable ML algorithms.	Compared multiple supervised ML classifiers for CKD prediction.	The study demonstrated strong predictive performance from machine-learning classifiers on CKD data.	Validate models on larger and more diverse datasets.
3	Ghosh & Khandoker (2024)	Investigation on explainable machine learning models to predict chronic kidney diseases	Scientific Reports, 14, 3687. DOI: 10.1038/s41598-024-54375-4	Prediction models need greater interpretability for healthcare applications.	Investigated explainable ML models for CKD prediction.	Explainability can help interpret which clinical variables contribute to predictions.	Further investigate explainability and external clinical validation.
4	Dharmarathne et al. (2024)	On the diagnosis of chronic kidney disease using a machine learning-based interface with explainable artificial intelligence	Intelligent Systems with Applications, 22, 200397. DOI: 10.1016/j.iswa.2024.200397	Need practical CKD diagnosis support while retaining model interpretability.	Evaluated several ML classifiers including Decision Tree, KNN, SVM, Random Forest, XGBoost and ANN with explainable AI.	Different classifiers can provide useful CKD prediction, while XAI improves interpretability.	Test the approach on broader patient populations and real clinical settings.
5	Gupta et al. (2026)	Data-driven explainable chronic kidney disease detection using RF based data imputation and meta-ensemble learning	Scientific Reports, 16, 12679. DOI: 10.1038/s41598-026-41425-2	Missing values and model reliability are important challenges in CKD prediction.	Used Random Forest-based imputation and a meta-ensemble learning approach with explainability.	The study reported strong CKD detection performance and showed the usefulness of ensemble methods.	Evaluate robustness using larger independent datasets and prospective validation.

3. Methodology

3.1 Proposed System Flow Diagram

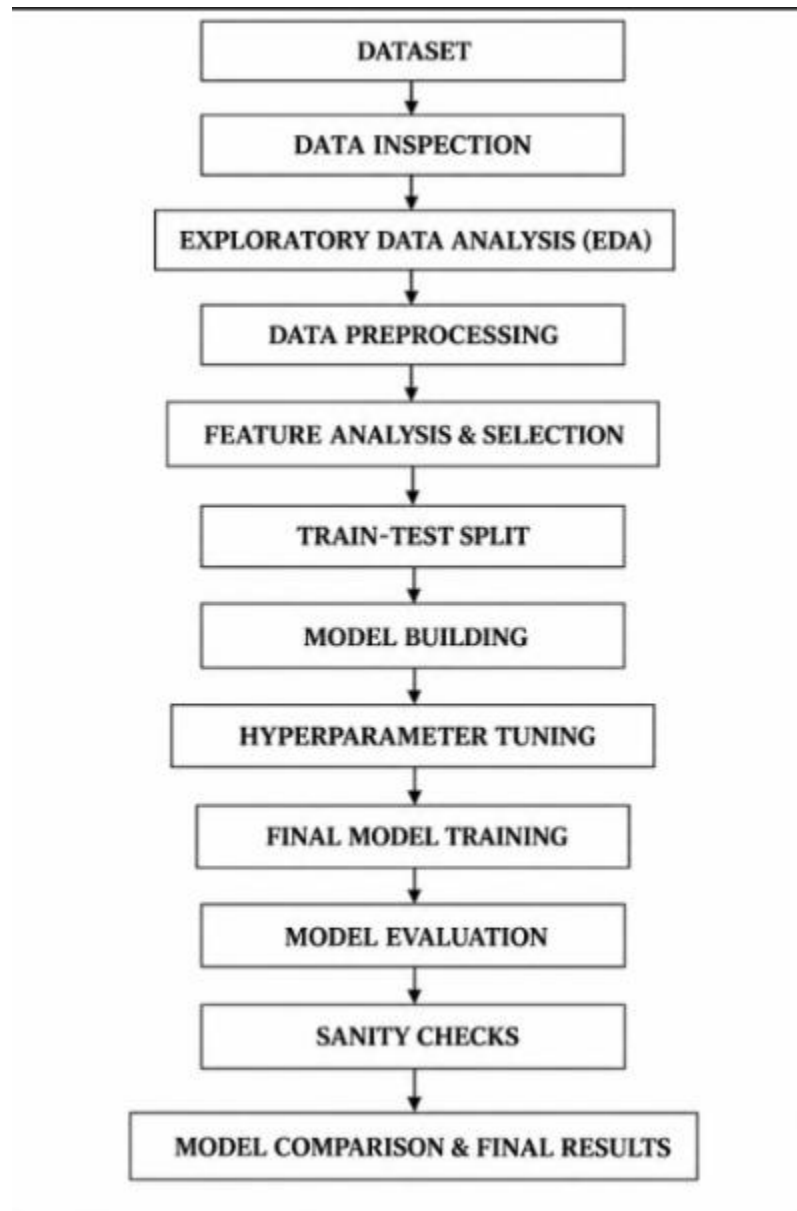


Figure Proposed CKD machine learning workflow.

The workflow follows the dependency order: dataset inspection, data cleaning and missing-value treatment, encoding and scaling, feature analysis, stratified splitting, model training, hyperparameter tuning, prediction, evaluation, and comparison.

3.2 Dataset Description

The project uses the Chronic Kidney Disease dataset from the UCI Machine Learning Repository (dataset 336). It contains 400 patient records, 24 predictor attributes, and a binary class target, giving 25 columns in the working CSV. The target classes are ckd and notckd. The UCI repository identifies 250 CKD and 150 not-CKD records and reports missing values. Predictor types include numerical, nominal, and binary clinical variables such as age, blood pressure, specific gravity, albumin, blood glucose, blood urea, serum creatinine, sodium, potassium,

hemoglobin, packed cell volume, white blood cell count, red blood cell count, hypertension, diabetes mellitus, coronary artery disease, appetite, pedal edema, and anemia.

Dataset citation: Rubini, L., Soundarapandian, P., & Eswaran, P. (2015). Chronic Kidney Disease [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5G020.

3.3 Experimental Setup

Component	Configuration
Operating System	Windows
Python	3.14.3
Environment	CKD.venv
IDE/Notebook	VS Code / Jupyter
Libraries	Pandas, NumPy, Scikit-learn, Matplotlib, XGBoost, LightGBM
Dataset	400 × 25
Split	80:20 stratified; 320 train / 80 test; random_state=42
CV	5-fold Stratified CV during tuning

3.4 Data Preprocessing

- Schema and data types were inspected before modeling.
- The original raw CSV was preserved unchanged.
- Exact duplicate rows were checked; none were found.
- Numerical predictors used median imputation followed by StandardScaler.
- Categorical predictors used most-frequent imputation followed by OneHotEncoder(handle_unknown='ignore').
- The target was mapped as ckd=1 and notckd=0.
- Preprocessing was fitted only on training data to prevent test-set leakage.

3.5 Exploratory Data Analysis

EDA examined descriptive statistics, target balance, missingness, feature distributions, categorical frequencies, and relationships between clinical variables. The target distribution is 250 CKD and 150 not-CKD cases (62.5% and 37.5%). Missing values occur across several clinical attributes, making systematic imputation necessary. Variables such as serum creatinine, hemoglobin, packed cell volume, albumin, specific gravity, blood urea, hypertension and diabetes mellitus are particularly relevant to the modeling analysis.

3.6 Feature Engineering and Feature Selection

A hybrid filter-plus-embedded feature analysis was performed using Mutual Information and Random Forest Mean Decrease in Impurity. The combined ranking identified the following top 12 features: hemo, pc, pcv, sc, rbc, sg, rbcc, htn, al, dm, sod, and bu. The selection procedure was designed to use training data to reduce leakage. The current Step 7 benchmark retained all 24 predictors, so the reported test results correspond to the full-feature pipeline. The selected 12-feature subset is retained for a controlled follow-up experiment.

3.7 Model Building and Training

- Logistic Regression (baseline)
- Decision Tree
- K-Nearest Neighbors
- Support Vector Machine
- Random Forest
- AdaBoost
- Gradient Boosting

- XGBoost
- LightGBM

These models represent linear, distance-based, kernel, tree, bagging, and boosting approaches. Conservative initial settings were used before tuning to control unnecessary model complexity.

3.8 Hyperparameter Tuning

RandomizedSearchCV was selected instead of exhaustive GridSearchCV to reduce computational cost on the 400-record dataset. Tuning used stratified 5-fold cross-validation on training data and F1-score as the optimization metric. Search spaces included model-specific regularization, depth, leaf size, estimator count, learning rate, subsampling and related parameters. A controlled three-iteration experiment was used to validate the end-to-end tuning process across all nine models. The held-out test set was not used for hyperparameter selection.

3.9 Model Evaluation

The models were evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrices and classification reports. Recall is particularly important for CKD screening because false-negative cases represent patients whose CKD status is not detected by the classifier.

4. Results and Discussion

4.1 Preprocessing Results

Stage	Records	Dimensions	Observation
Raw dataset	400	25 columns	24 predictors + target
Raw missing cells	1012	—	Missing values present
Exact duplicates	0	—	No exact duplicates
Training set	320	24 predictors	Stratified
Test set	80	24 predictors	Stratified

The preprocessing pipeline handles missing values and mixed data types without overwriting the original raw dataset. The held-out test set contains 80 records, with 30 not-CKD and 50 CKD cases.

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
LightGBM	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000
Gradient Boosting	1.0000	1.0000	1.0000	1.0000	1.0000
AdaBoost	1.0000	1.0000	1.0000	1.0000	1.0000
Decision Tree	0.9875	1.0000	0.9800	0.9899	0.9900
Logistic Regression	0.9750	1.0000	0.9600	0.9796	0.9933
SVM	0.9750	1.0000	0.9600	0.9796	1.0000
KNN	0.9625	1.0000	0.9400	0.9691	0.9900

On the fixed 80-record test split, Random Forest, LightGBM, XGBoost, Gradient Boosting and AdaBoost each achieved 100% across the reported metrics. Decision Tree achieved 98.75% accuracy, while Logistic Regression and SVM achieved 97.50% and KNN achieved 96.25%. The results are strong benchmark outcomes, but repeated perfect scores on a small test set should be interpreted cautiously.

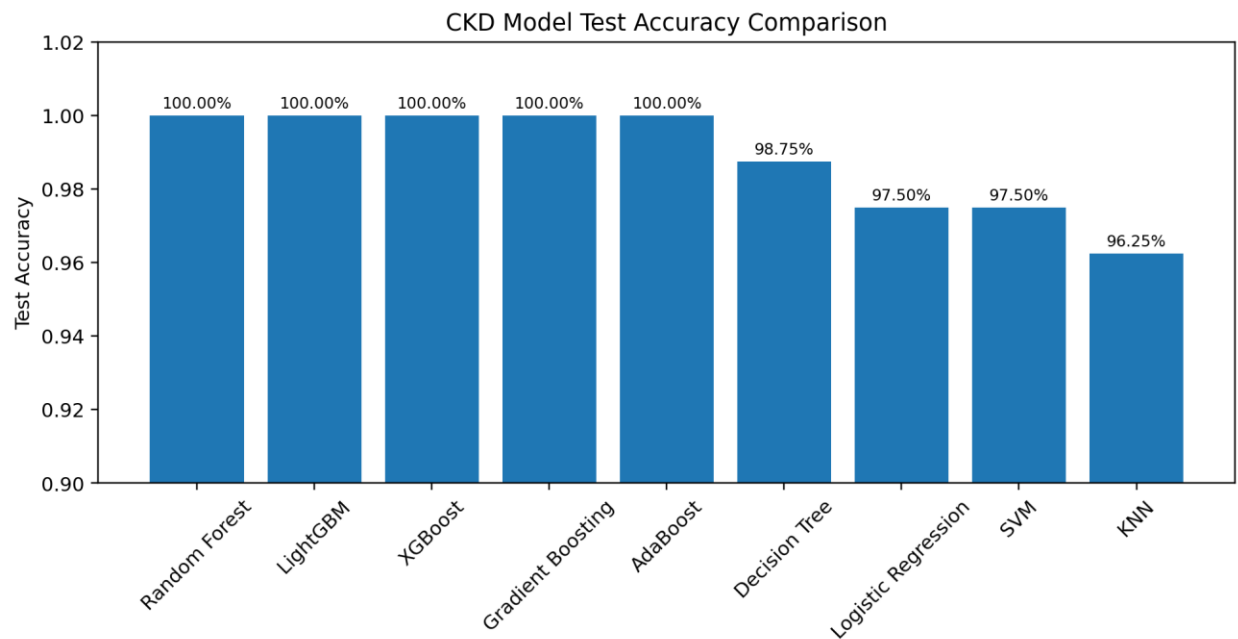


Figure 2. Test accuracy comparison across the nine evaluated models.

4.3 Visual Results and Discussion

The comparison indicates that the ensemble and boosting models produced the strongest observed test-set metrics, while the baseline and distance-based approaches had slightly lower recall. Tree-based ensemble methods can capture nonlinear interactions among heterogeneous clinical variables, which may contribute to their strong benchmark performance. However, the observed results should not be interpreted as evidence of clinical superiority or perfect real-world performance.

The sanity-check stage found no exact duplicate records, no direct target-column leakage, and no exact feature-row overlap between the recreated training and test partitions. These checks reduce concern about obvious structural contamination but do not rule out every form of dataset-specific optimism or external-validity limitation. Cross-validation results from tuning should therefore be interpreted together with the final held-out test results.

Recall is particularly relevant for a screening-oriented CKD classifier. The five models with perfect test performance achieved recall of 1.00; Decision Tree achieved 0.98, Logistic Regression and SVM 0.96, and KNN 0.94. Confusion matrices should be included in the final presentation to show the exact false-positive and false-negative counts.

5. Conclusion

This project developed a reproducible hybrid machine learning framework for CKD classification using the UCI Chronic Kidney Disease dataset. The workflow covered data validation, missing-value treatment, encoding, scaling, exploratory analysis, feature selection, model training, randomized hyperparameter tuning, sanity checking and evaluation. Nine algorithms from multiple model families were compared. On the current fixed 80-record test split, five models achieved 100% on the reported metrics, while the remaining models also achieved high performance. The study demonstrates the feasibility of machine learning for this benchmark classification task, but the results should be treated as academic benchmark evidence rather than clinical validation because the dataset contains only 400 records and does not represent the diversity of real-world clinical populations.

6. Future Work

- Validate the framework on larger and more diverse clinical datasets.
- Compare the full 24-feature pipeline with the selected 12-feature subset.
- Perform external validation and probability calibration.
- Add SHAP/LIME explanations for patient-level interpretability.
- Investigate soft-voting or stacking ensembles after individual-model validation is finalized.
- Develop a web-based CKD prediction and model-results dashboard.
- Study model drift and robustness under distribution changes.
- Conduct clinical usability and prospective validation before any real-world deployment.

7. References

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